

LAPALU LIYANAGE

Software Engineering Undergraduate | Full-Stack Developer

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OBJECTIVE

Third-year Software Engineering undergraduate seeking a Software Engineering Internship where I can apply and grow my full-stack skills. Hands-on project experience across Java/Spring Boot, Python/Flask, and React — including designing a wallet/payment-style backend with JWT authentication and PostgreSQL. Comfortable working end-to-end, from schema design to deployment, and eager to learn in a real engineering team.

EDUCATION

Reading for BSc in Information Technology

Rajarata University of Sri Lanka (RUSL), Mihinthale — 3rd Year

Saranath National College, Kuliyaipitiya

G.C.E. Advanced Level — 2020 | G.C.E. Ordinary Level — 2017

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, C++
- **Backend:** Spring Boot, Spring Security, Flask, Node/Express, REST API design, JWT authentication
- **Frontend & Mobile:** React (Vite), Context API, Android (Java, MVVM, Room, WorkManager)
- **Databases:** PostgreSQL, MySQL, Supabase, JPA/Hibernate
- **Tools & DevOps:** Git/GitHub, Docker, Postman, pgAdmin, Render, Vercel, GitHub Actions
- **ML/AI (exposure):** Google Gemini API integration, scikit-learn, sentence-transformers, basic NLP-based scoring models

KEY PROJECTS

CeylonPay — Wallet & P2P Payment API — *Java 21, Spring Boot 4, PostgreSQL, JWT, React, Docker* [\(repo\)](#)

- Built a full wallet system as a personal deep-dive into fintech-style backend engineering: account registration, deposits, and peer-to-peer transfers by phone number.
- Designed a four-table schema (users, wallets, transactions, audit_log) with atomic, @Transactional transfer logic so balances and audit records never go out of sync.
- Deployed a live demo: React frontend on Vercel, Spring Boot API + Postgres on Render, containerized with a multi-stage Docker build.

Enhanced Ant Colony Algorithm — *Python, NumPy, Pandas* — *Multi-Objective Optimization* [\(repo\)](#)

- An ant-colony-inspired heuristic for the Multi-Objective Transportation Problem, building a fixed pheromone-like probability matrix from combined objective cost matrices.
- Implemented and benchmarked four cost-combining strategies (geometric, arithmetic, harmonic mean, and minimum) against a Vogel's-Approximation-style greedy allocator.

MediBloom — Medication Management App — *Java, Android (MVVM), Room, Retrofit, Google Gemini API, WorkManager*

- Android app for tracking medicines, dose reminders, and adherence, with dose logging, a calendar view, mood tracking, and a Gemini-powered AI health-assistant chat.
- Layered data/domain/ui architecture (MVVM with LiveData) backed by a local, offline-first Room database — no cloud account required.
- Exact-alarm reminders that reschedule automatically after device reboot, with notification actions to mark a dose taken directly from the tray.

AI-Enhanced Smart Parking System — *Java 17, Spring Boot 3, PostgreSQL, JWT, Google Gemini API, JUnit/Testcontainers* [\(repo\)](#)

- Backend for booking parking slots with real-time availability, JWT-secured reservations, automatic expiry, and dynamic pricing.
- Added an AI layer (Google Gemini) for demand prediction and a natural-language booking assistant, isolated behind an AiAssistantService interface so the test suite never depends on a live API key.
- Solved concurrent-booking race conditions with a pessimistic row lock inside a single transaction, verified by a dedicated concurrency test firing 10 simultaneous requests at one slot.

DevScore (ScriptFusion) — Automated Job-Readiness Scoring — *Node/Express, React, Python/scikit-learn, Supabase*

- Final-year research project (6-person team) that verifies resume-claimed skills against real GitHub activity to produce a Weighted Verification Ratio (WVR), combining semantic skill matching (sentence-transformers) with AST-based code analysis.
- Co-authored an IEEE-format conference paper documenting the system and methodology, validated against ~48 processed candidate profiles.

ADDITIONAL PROJECTS

Facebook Album Download Kit (Full-Frame) — Chrome/Edge extension (JS, Manifest V3) to download a user's own Facebook albums as files or a ZIP, with a dependency-free ZIP writer and Node test suite.

SmartPlanner — Personal Organizer App — Object-oriented personal organizer/planner application (C++) built as coursework, structured around encapsulated task and schedule classes.

ChordScope — Flask/React app that identifies advanced chords (7ths, extensions, sus, inversions) from a YouTube link's audio via a 144-chord template-matching recognizer.

CERTIFICATIONS

- **Introduction to Software Engineering** — IBM, via Coursera (Aug 2024)
- **Front-End Development, Programming with JavaScript, Version Control** — Meta, via Coursera (Dec 2023 – May 2024)

LEADERSHIP & EXTRACURRICULAR

- **President, Fastunes Music Society**, Faculty of Applied Sciences, RUSL
- Lead a student society's administration and events — organizing an online audition drive, coordinating the 2026 event calendar, and handling formal society correspondence.
- **Secretary, Science Club**, Saranath National College, Kuliyaipitiya (2019)
- **InnovateX 2025 Grand Finale** — represented Team LoopCode from Rajarata University of Sri Lanka, showcasing innovation, problem-solving, and teamwork skills.